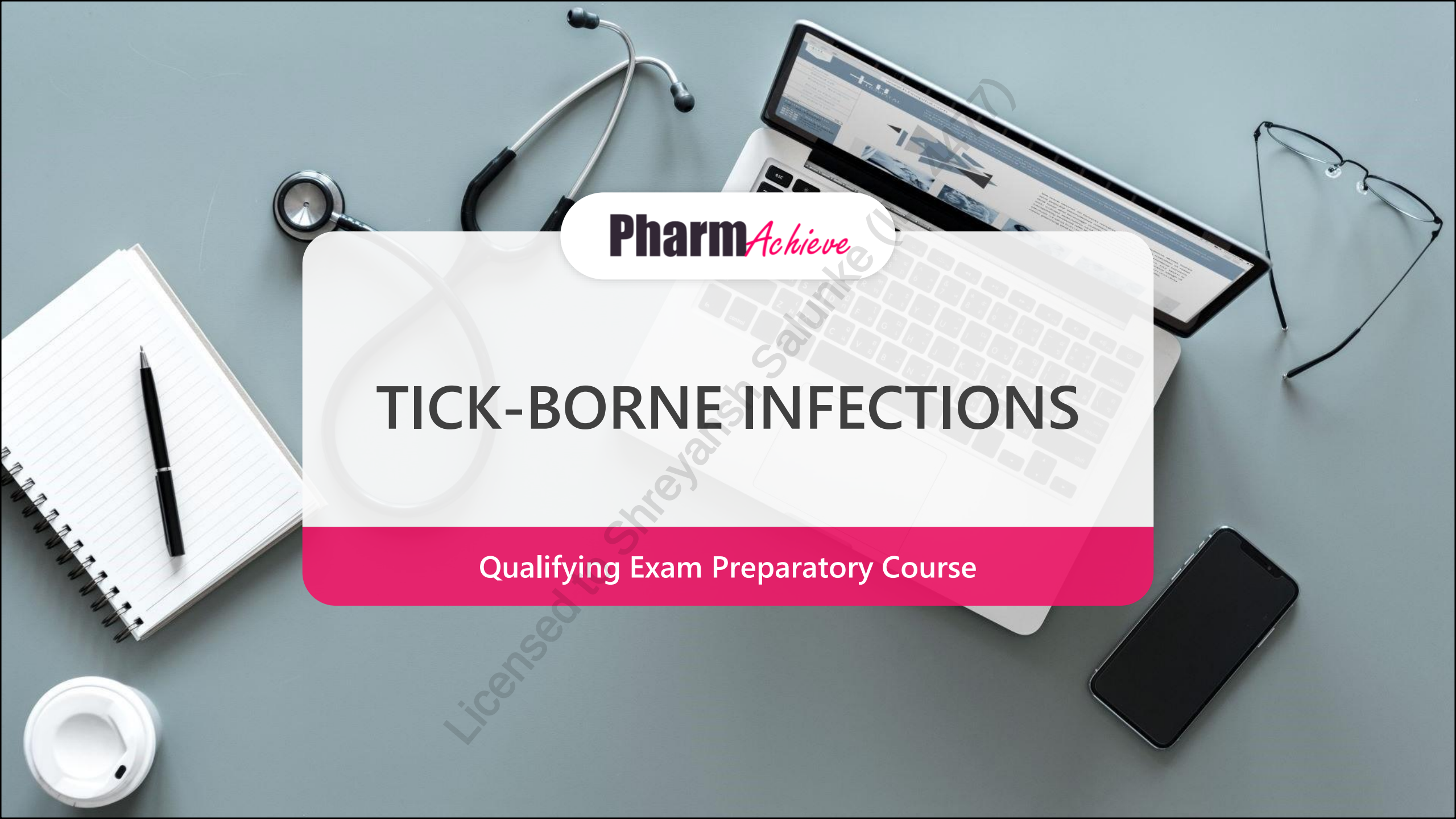




Pharm*Achieve*

TICK-BORNE INFECTIONS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

DEET Diethyltoluamide / N,N-Diethyl-Meta-Toluamide

EM Erythema Migrans

B. burgdorferi *Borrelia Burgdorferi*

IDSA Infectious Diseases Society of America

IV Intravenous

NSAIDs Nonsteroidal Anti-inflammatory Drugs

PCR Polymerase Chain Reaction

PHAC Public Health Agency of Canada

PTLDS Post-Treatment Lyme Disease Syndrome

RMSF Rocky Mountain Spotted Fever

STARI Southern Tick-Associated Rash Illness

US United States

TYPES OF TICKS AND TICK-BORNE DISEASES

Common Name	Latin Name	Endemic areas	Associated Disease
Blacklegged Tick	<i>Ixodes scapularis</i>	Eastern Canada/US	Lyme, Babesiosis, Anaplasmosis, Powassan, Borrelia
Lone Start Tick	<i>Amblyomma americanum</i>	Eastern Canada/ Southeastern US	Ehrlichiosis, tularemia, Heartland, Bourbon, STARI
Black Dog Tick	<i>Dermacentor variabilis</i>	Canada, west coast/Central US	Tularemia, RMSF
Brown Dog Tick	<i>Rhipicephalus sanguineus</i>	Southwestern US (indoor environments with dogs)	RMSF
Groundhog Tick	<i>Ixodes cookei</i>	Central-Eastern Canada/Eastern US	Powassan
Rocky Mountain Tick	<i>Dermacentor andersoni</i>	Eastern Canada/ Central/Mountain US	RMSF, tularemia
Western Blacklegged Tick	<i>Ixodes pacificus</i>	Canada/Western US	Lyme, anaplasmosis

DEFINITIONS TICKBORNE INFECTIONS

- **Ticks**

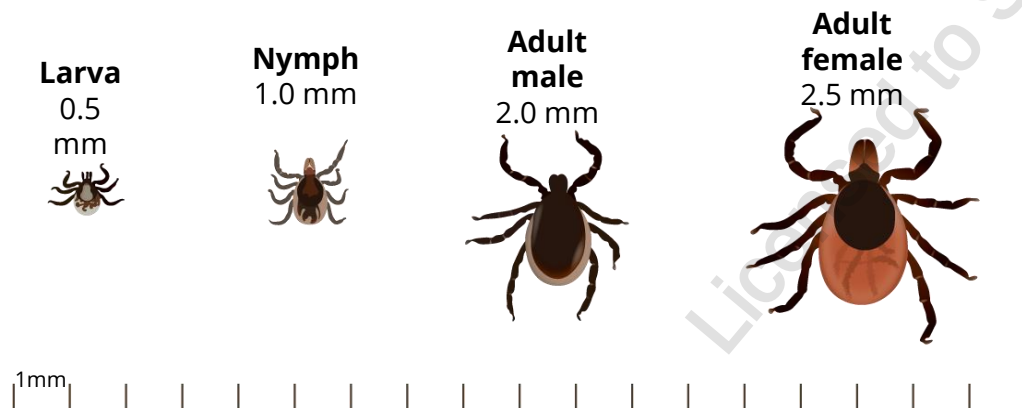
- Hemophagocytic ectoparasitic arachnids
- Family: Ixodidae

- **Distribution**

- Worldwide (>in temperate climates)

- **Food source:**

- Blood: mammal, bird and others



- **Tickborne diseases:**

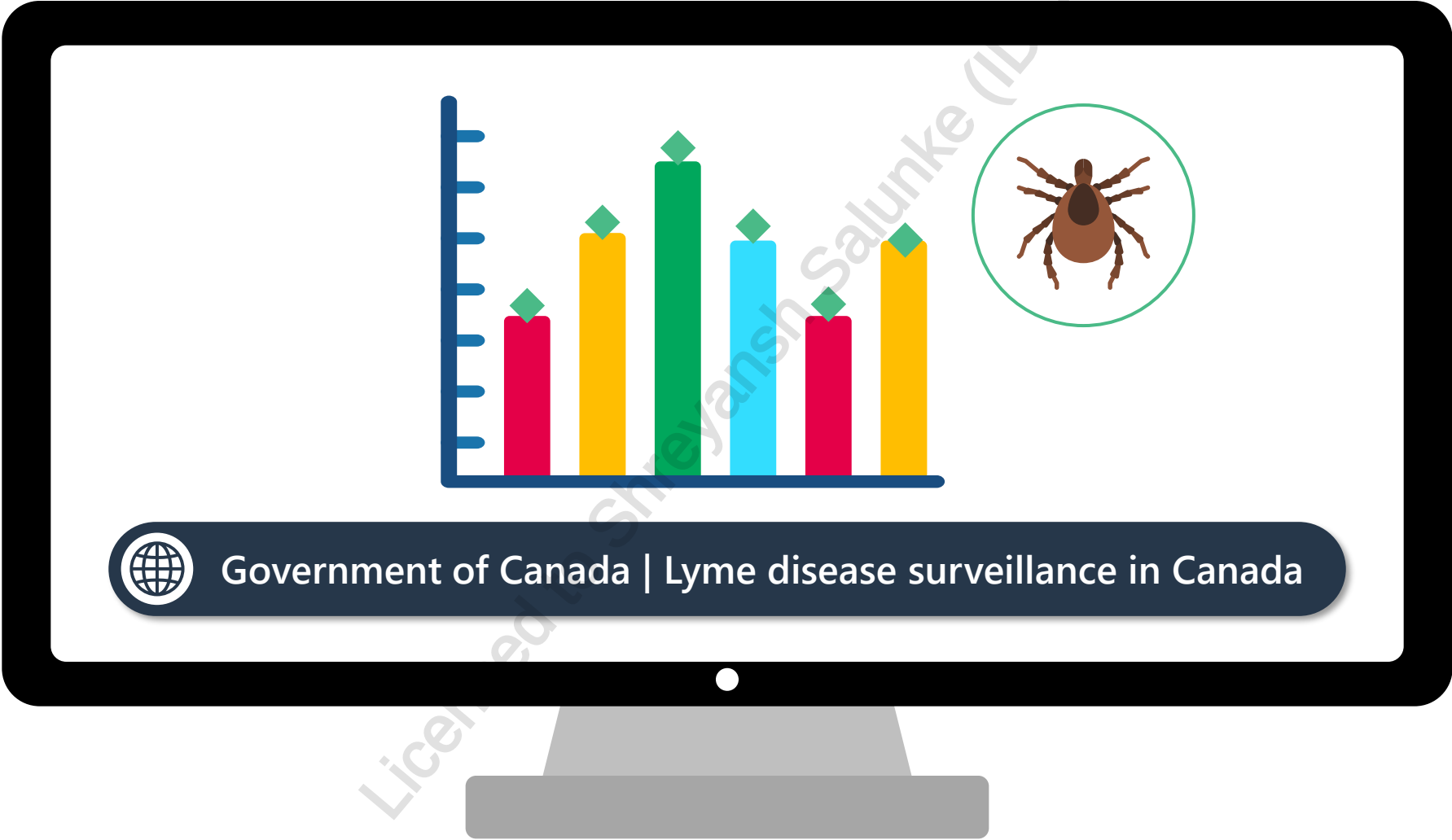
- Anaplasmosis
- Babesiosis
- Borrelia
- Colorado Tick Fever
- Ehrlichiosis
- Heartland Virus
- Lyme Disease
- Powassan Disease
- Rickettsia
- Rocky Mountain Spotted Fever
- STARI
- Tickborne Relapsing Fever
- Tularemia

LYME DISEASE

- Organism:
 - *B. burgdorferi*:
 - Spirochete
- Vector:
 - *I. scalpularis*
 - *I. pacificus*
- Identification:
 - Dark field microscopy (difficult/impractical)
 - PCR
 - Antibody screen

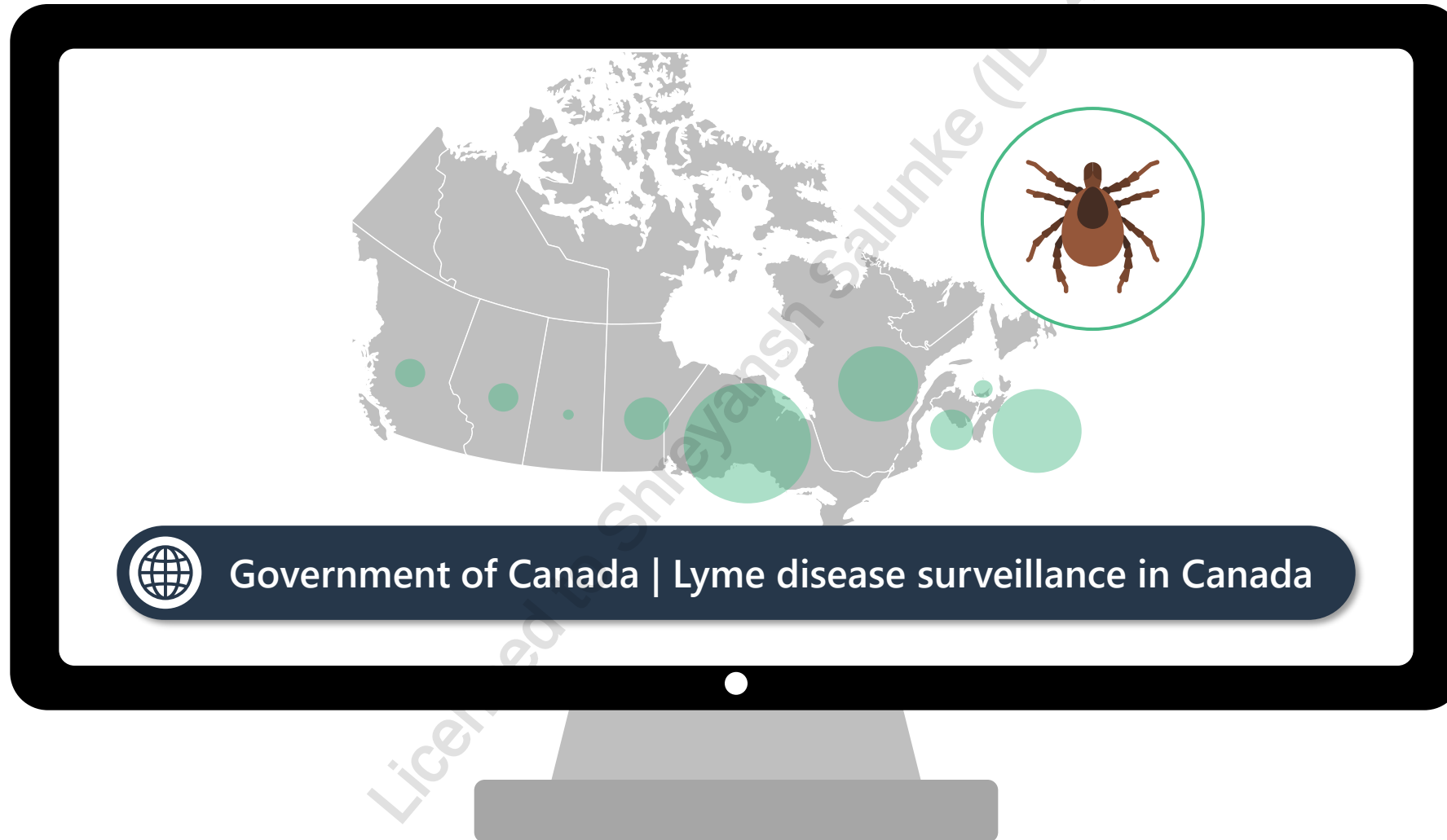


INCIDENCE



GEOGRAPHIC LOCALIZATION

FYI



RISK FACTORS

- Outdoor exposure in endemic areas
- Clothing that leaves exposed skin
- Failure to use insect repellent
- Pets that can bring ticks into the home/yard

LYME DISEASE

- **Presentation:**

- Early localized erythema migrans (days to weeks)
 - Rash: homogenously erythematous OR central clearing with "bullseye" pattern
 - Appears within 7 to 10 days
 - Size ranges from 5 to 60 cm
 - Non-pruritic and painless
 - Low-grade fever*, fatigue
- Early, disseminated (less than 3 months):
 - Vague, includes muscle aches, fatigue, encephalitis/meningitis, cardiac conduction issues
- Late/Persistent (more than 3 months):
 - Neurologic: confusion, fatigue, encephalopathy
 - Rheumatologic: multiple joint arthritis

- **Diagnosis:**

- Clinical symptoms/history
- PCR and/or antibody serology screening

- **Prognosis:**

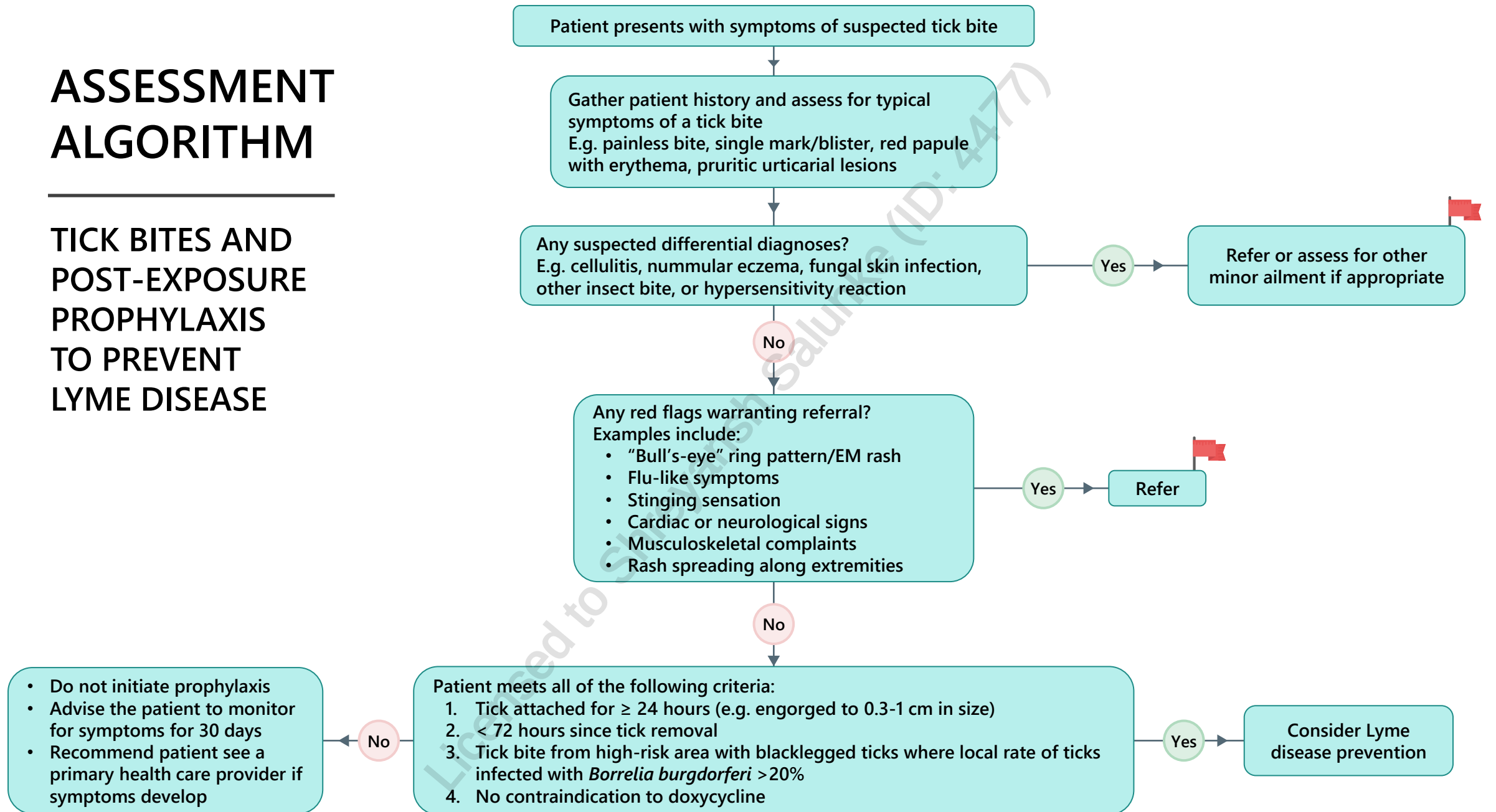
- Dependent on treatment, residual symptoms >6 months possible in disseminated disease = Post-Treatment Lyme Disease Syndrome (PTLDS)



*Fever can appear in the later stages (e.g. a symptom of meningitis)

ASSESSMENT ALGORITHM

TICK BITES AND POST-EXPOSURE PROPHYLAXIS TO PREVENT LYME DISEASE



GOALS OF THERAPY

- ✓ Prevent Lyme Disease/other tick-borne infections
- ✓ Treat infections to prevent:
 - ✓ Acute decompensation
 - ✓ Morbidity (short-term/long-term)
 - ✓ Mortality
- ✓ Avoid adverse effects of medications

TREATMENT ALGORITHM

TICK BITES AND POST-EXPOSURE PROPHYLAXIS TO PREVENT LYME DISEASE

Treatment indicated for suspected tick bite and post-exposure prophylaxis to prevent Lyme disease

Counsel on prevention and non-pharmacological options:

- *Pre-exposure prevention strategies*
 - Wear light-coloured clothing to easily spot ticks
 - Wear closed toe shoes and long-sleeved shirts and pants to provide barriers to ticks (clothing should be tucked in to minimize exposure)
 - Use insect repellent (e.g. DEET, icaridin)
- *Post-exposure prevention strategies*
 - Inspect clothing, pets, and body for any more ticks
 - Remove tick(s) and clean the affected areas
 - Shower or bathe within 2 hours after returning from outdoors
 - Put clothes in dryer on high heat to kill ticks on clothing

Pharmacological treatment options:

- Doxycycline single dose to be initiated within 72 hours of tick removal

Continue monitoring for Lyme disease symptoms for at least 30 days

Have any Lyme disease symptoms occurred?

No

Educate the patient that late Lyme disease symptoms can occur years after initial infection, and refer if suspected

Yes

Refer

NON-DRUG MEASURES: PREVENTION

Pre-exposure prevention

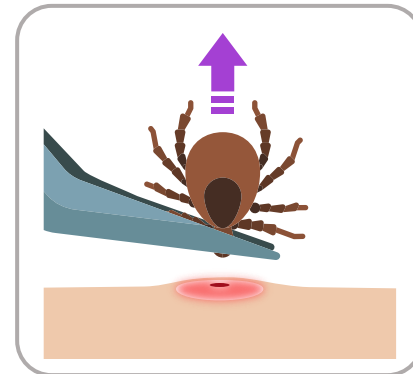
- Wear closed-toe shoes, long-sleeved shirts and pants
- Wear light-coloured clothes to spot ticks easier
- Pull socks over pant legs
- DEET-containing insect repellent (up to 30% for > 12 years of age)

Post-exposure prevention

- “Tick check” after exposure to endemic areas
- Rapid removal of any attached ticks with tweezers (do NOT burn)
- Shower or bathe within 2 hours of being outdoors to wash away loose ticks

NON-DRUG MEASURES: TICK REMOVAL

- Use fine-tipped tweezers to grasp the tick as close to the skin's surface as possible
- Pull upward with steady, even pressure. Don't twist or jerk the tick; this can cause the mouth-parts to break off and remain in the skin. If this happens, remove the mouth-parts with tweezers. If you are unable to remove the mouth easily with clean tweezers, leave it alone and let the skin heal
- After removing the tick, thoroughly clean the bite area and your hands with rubbing alcohol or soap and water
- Never crush a tick with your fingers. Dispose of a live tick by putting it in alcohol, placing it in a sealed bag/container, wrapping it tightly in tape, or flushing it down the toilet



DRUG THERAPY

- **Post-exposure prophylaxis given within 72 hours of a HIGH-RISK tick bite**
High risk: 1) Ixodes spp tick, 2) Highly endemic area, 3) Attached >24 hours
 - Doxycycline:
 - Adults: 200 mg PO x 1 dose
 - Children: < 45 kg: 4 mg/kg PO x 1 dose
 - Adverse effects: nausea, esophagitis, abdominal pain
 - Avoid antacids, dairy products, and take on an empty stomach for optimal absorption
- **Erythema Migrans (early disease, updated 2020 IDSA guidelines)**
 - Adults:
 - Doxycycline x 10 days
 - Cefuroxime x 14 days
 - Amoxicillin x 14 days
 - Children (< 8 years old)
 - Amoxicillin x 14 days
 - Cefuroxime x 14 days
 - Doxycycline x 10 days (case by case)
 - Azithromycin x 7 days is 2nd line when other therapies are contraindicated or not tolerated

DRUG THERAPY: SPECIAL POPULATIONS

- **Children**

- All ages may receive post-exposure preventative doxycycline single dose
- Insect repellent
 - > 6 months: DEET or icaridin
 - < 6 months: soybean oil, lasts ~3.5h
 - < 2 years-old: avoid the use of citronella or lavender oils due to lack of safety evidence

- **Pregnancy/Breastfeeding**

- Safe to use post-exposure preventative doxycycline single dose
- Insect repellent
 - Avoid chemical repellents, if possible
 - If repellent is needed: use low-strength product sparingly, wash off after protection no longer needed
 - Non-pharm methods: long clothing, avoidance of insect habitats

DRUG THERAPY: TREATMENT OF LATE OR DISSEMINATED LYME DISEASE

Syndrome	Drug Choice	Duration (days)
Nervous System Involvement	Doxycycline OR Ceftriaxone	14-21
Carditis	Doxycycline, Amoxicillin, Cefuroxime or Ceftriaxone	14-21
Arthritis (initial/recurrent)	Doxycycline, Amoxicillin, Cefuroxime	28
Arthritis, recurrent (IV)	Ceftriaxone	14
Acrodermatitis chronica atrophicans	Doxycycline, Amoxicillin, Cefuroxime	21-28
Borrelial lymphocytoma	Doxycycline, Amoxicillin, Cefuroxime	14

DRUG THERAPY: ADJUNCTIVE TREATMENTS

Relief of muscle pain (myalgia) and headache

- NSAIDS
- Acetaminophen

Supportive care for chronic Lyme

- Occupational therapy
- Psychology/psychiatry

OTHER TICKBORNE INFECTIONS

FYI

Babesiosis

- Caused by *B. microti*, Vector is *I. scapularis*
- Causes hemolysis (severe in immunocompromised)
 - Asymptomatic: no treatment
 - Mild: atovaquone and azithromycin
 - Moderate/Severe: clindamycin and quinine

Anaplasmosis

- Caused by *A. phagocytophilum*. Vector is *I. scapularis*/*I. pacificus*.
- Causes flu-like syndrome with fever, leukopenia and thrombocytopenia
- Treated with doxycycline x 7-14 days (10 days if early Lyme coinfection)

OTHER TICKBORNE INFECTIONS

Ehrlichiosis

FYI

- Caused by *E. chaffeensis*. Vector is *A. americanum*
- Rash, fever, thrombocytopenia, leukopenia
- Treatment with doxycycline x 7-14 days

Rocky mountain spotted fever

- Caused by *R. rickettsii*, vector is *D. andersoni*
- Possibly severe/fatal disease in Southern Alberta/Saskatchewan
- Early treatment for adults and children with doxycycline

Tularemia

FYI

- Caused by *F. tularensis*, vector is *D. variabilis*
- Various and severe disease possible

MONITORING PARAMETERS (TICK BITE)

Ongoing monitoring of symptoms for 30 days

EFFICACY ENDPOINTS	
Parameter	Timeframe
Early localized symptoms (erythema migrans)	7-10 days
Early disseminated symptoms (e.g. muscle aches, fever, stiff neck, heart palpitations, chest pain)	Weeks- months (≤ 3 months)
Late/persistent symptoms (e.g. confusion, impaired memory, arthritis)	≥ 3 months

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Adverse effects of prophylactic therapy (GI upset, photosensitivity)	Prevent or minimize	48 hours

MONITORING PARAMETERS (CONFIRMED LYME DISEASE)

EFFICACY ENDPOINTS		
Parameter	Degree of change	Timeframe
Early localized symptoms (erythema migrans)	Improvement/ Complete recovery	Duration of therapy
Cardiac signs and symptoms (carditis)		1-6 weeks
Arthritis		Several months
Neurologic manifestations		Several months

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Adverse effects of therapy (GI upset, photosensitivity)	Prevent or minimize	Duration of therapy

TAKEAWAYS

- Educate patients about the importance of tick bite prevention strategies
- Ensure rapid removal of any attached tick (will reduce transmission of disease)
- Consider prophylaxis in endemic areas for Lyme disease
- Note that symptom resolution may take several months, especially in late disease
- Be aware of less common tick-borne infections in select patients

CASE 1

EM, a 32-year-old pregnant patient (14 weeks along), presents to the pharmacy after hiking in Algonquin Park (Ontario, endemic area) for 2 days and noticed a tick on her body yesterday. She took a picture of the tick before removing it, and you can tell it is engorged. She tells you it has been about 24 hours since she noticed it and removed it. She has no known medication allergies, and she has no medical conditions. She takes a daily prenatal vitamin, which contains about 35 mg of elemental iron.

1. Is EM eligible for post-exposure prophylaxis and if so, what treatment should you choose?
 - a) No, the tick has been on longer than 72 hours
 - b) Yes, a single dose of doxycycline
 - c) Yes, amoxicillin for 14 days because she is pregnant
 - d) No, she is pregnant and needs referral



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2. Which of the following are not important counselling point for EM, assuming you gave them doxycycline?
- a) Take with plenty of water and remain upright for at least 30 minutes
 - b) Take your prenatal vitamin at least 2 hours after your dose
 - c) Doxycycline can cause heartburn, to prevent this, you can take it with TUMS
 - d) It is preferable to take doxycycline on an empty stomach, avoiding eating for 2 hours after the dose



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3. In the case where EM did not qualify for post-exposure prophylaxis, but presented within 2 weeks with mild fever, muscle aches, and multiple, small EM lesions, what treatment is appropriate?
- a) Doxycycline x 10 days
 - b) Amoxicillin x 21 days
 - c) Cefuroxime x 14 days
 - d) Azithromycin x 7 days



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4. For future hikes, EM asks you what they can do to prevent tick bites, what should you NOT tell them?
- a) Wear dark-colored, long-sleeved clothing
 - b) Use DEET-based insect repellants
 - c) Shower immediately after being outside in endemic area
 - d) Remove ticks as soon as you discover them



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7. Wong KH, Shapiro ED, Soffer GK. A Review of Post-treatment Lyme Disease Syndrome and Chronic Lyme Disease for the Practicing Immunologist. *Clin Rev Allergy Immunol*. 2022;62(1):264-271. doi:10.1007/s12016-021-08906-w.

CHANGE LOG

March 2025

- Slide 3: Added legend
- Slide 4: Italicized species, corrected endemic areas and diseases
- Slide 6: Italicized species names
- Slide 11: Re-defined the rash
- Slide 15: Changed to 'up to 30% DEET', removed 'save the tick'
- Slide 20: Reformatted graphic
- Slide 21: Italicized species names
- Slide 29: Added references

July 2025

- Content reviewed; no changes were made

September/October 2025

- Slide 3: Updated legend
- Slide 11: Minor edits in clinical presentation, added note on fever, and note on PTLDS
- Slide 23-24: Added monitoring parameters slides
- Slide 30: Updated references

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 16: Minor update
- Added FYI to slide 20 and parts of slide 21
- Slide 22: Minor update to monitoring parameters